

Assignment 2

Use the following questions to guide your understanding of these topics. You do not need to answer these questions directly in the paragraph.

Part 1: Importance Sampling (1 paragraph, ~250 words)

1. What type of problem does importance sampling help solve? What kind of integral or expectation are we trying to compute? Why might it be difficult to compute directly?
2. What is the core idea of importance sampling? What is the “proposal distribution”? Why do we sample from a different distribution than the one we care about?
3. What role do importance weights play? What do they correct for? How do they allow us to approximate expectations?
4. When can importance sampling perform poorly? What happens if the proposal distribution is poorly chosen?

Part 2: Variational Inference (1 paragraph, ~250 words)

1. What problem is variational inference trying to solve? What is a posterior distribution? Why is it often intractable?
2. What is the main idea behind variational inference? What is the variational family? What does it mean to “approximate” a posterior?
3. How does optimization enter the picture? What is being minimized or maximized? Why is KL divergence important?